

Supplementary File 4. PRISMA-ScR Checklist

Review Title

Reframing the Malaria-Sickle Cell Disease Relationship in Sub-Saharan Africa as a Bounded Syndemic: A Scoping Review and Evidence Gap Map

Purpose

This supplementary file reports how the completed scoping review addressed the PRISMA Extension for Scoping Reviews, PRISMA-ScR. The checklist identifies where each reporting item is addressed in the manuscript and supplementary files.

Completed PRISMA-ScR Checklist

Item	Section	PRISMA-ScR checklist item	Reported	Location in manuscript or supplementary files	Reporting details
1	Title	Identify the report as a scoping review.	Yes	Title page	The title identifies the manuscript as a scoping review of the malaria and sickle cell disease syndemic in sub-Saharan Africa.
2	Abstract	Provide a structured	Yes	Structured Abstract	The abstract summarises

		summary including background, objectives, eligibility criteria, sources of evidence, charting methods, results, and conclusions.			the rationale, review aim, PCC-based eligibility criteria, databases and grey literature sources, evidence charting, domain-based synthesis, cross-domain findings, and conclusion.
3	Introduction, rationale	Describe the rationale for the review in the context of what is already known and explain why a scoping review is appropriate.	Yes	Section 1, Introduction	The introduction explains the fragmented evidence base across malaria-HbS biology, HbS geography, SCD burden, severe anaemia,

					under-recognition, newborn screening, routine data, structural vulnerability , and programme response. It explains why a scoping review is appropriate for mapping heterogeneous evidence rather than estimating a pooled effect.
4	Introduction, objectives	Provide an explicit statement of review questions and objectives with reference to	Yes	Section 2, Review Aim and Questions; Section 4.3, Review Questions	The manuscript states the main review question and nine sub-questions. These address

		key elements such as population, concept, and context.			biological interaction, spatial co-risk, clinical convergence , service-mediated visibility, structural vulnerability , programme response, syndemic theory use, and evidence gaps in SSA.
5	Methods, protocol and registration	Indicate whether a review protocol exists, state where it can be accessed, and provide registration information if available.	Yes	Section 4.1, Review Design; Supplementary File 1	The review protocol is provided as Supplementary File 1. The manuscript reports that the review followed a pre-specified protocol informed by JBI scoping review

					methodology and PRISMA-ScR. Registry information should be inserted if the protocol is registered on OSF or another platform.
6	Methods, eligibility criteria	Specify characteristics of sources of evidence used as eligibility criteria, including years considered, language, publication status, and rationale.	Yes	Section 4.4, Eligibility Criteria; Section 4.5, Inclusion and Exclusion Criteria; Supplementary Files 1 and 3	Eligibility criteria are structured using Population, Concept, Context. The review included peer-reviewed and grey literature sources relevant to malaria, SCD, HbS, red blood

					cell polymorphisms, severe anaemia, service visibility, structural vulnerability, programme response, and syndemic theory in SSA. No date restriction was applied at the initial search stage. No language restriction was applied where feasible.
7	Methods, information sources	Describe all information sources used in the search, including databases,	Yes	Section 4.6, Information Sources; Supplementary File 2	The manuscript reports searches of PubMed or MEDLINE, Embase,

		grey literature sources, and citation searching, with the date of the most recent search.			Scopus, Web of Science Core Collection, Global Health, CINAHL, African Index Medicus, and Cochrane Library. Grey literature sources included WHO, Africa CDC, SickleInAfrica, ASH resources relevant to CONSA, ministries of health, GBD resources, institutional repositories, conference
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					proceedings, and SCD implementation networks. Search dates and record counts are documented in Supplementary File 2.
8	Methods, search	Present the full electronic search strategy for at least one database, including limits used, so that it can be repeated.	Yes	Section 4.7, Search Strategy; Supplementary File 2	The manuscript summarises the search concepts, while Supplementary File 2 provides full database-specific search strings for PubMed or MEDLINE, Embase, Scopus, Web of Science,

					Global Health, CINAHL, African Index Medicus, and Cochrane Library. Search limits and grey literature search terms are also reported.
9	Methods, selection of sources of evidence	State the process for selecting sources of evidence, including title and abstract screening, full-text screening, number of reviewers, and how disagreements	Yes	Section 4.8, Record Management and Study Selection; Supplementary File 3	The manuscript reports two-stage screening: title and abstract screening followed by full-text screening. At least two reviewers independently screened records.

		ts were resolved.			Disagreements were resolved through discussion and adjudication by the review lead or a third reviewer. The detailed screening guide is provided in Supplementary File 3.
10	Methods, data charting process	Describe the data charting process, including whether forms were piloted, number of reviewers, and how disagreements were resolved.	Yes	Section 4.9, Data Charting; Supplementary File 5	The manuscript reports that a standardised data charting form was piloted and refined before full extraction. Data charting captured bibliographi

					c, geographic, biological, clinical, health- system, programme, and syndemic coding variables. Ambiguous extraction decisions were resolved through team discussion.
11	Methods, data items	List and define all variables for which data were sought and any assumptions or simplificatio ns made.	Yes	Section 4.9, Data Charting; Supplementa ry File 5	Variables included author, year, country, study design, population, genotype or polymorphis m, malaria outcome, SCD

					outcome, severe anaemia, transfusion need, routine data, newborn screening, confirmatory diagnosis, referral, retention, structural vulnerability, programme response, syndemic theory use, and evidence gaps.
12	Methods, critical appraisal of individual sources of evidence	If done, provide rationale for appraising included sources, describe methods used, and explain how appraisal	Yes, with explanation	Section 4.12, Methodological Quality Appraisal	Formal critical appraisal was not used as an inclusion criterion because the review aimed to map the

		informed synthesis.			range, nature, and integration of evidence rather than pool effect estimates. Methodological characteristics were charted to support interpretation of evidence strength and limitations.
13	Methods, synthesis of results	Describe methods for handling and summarising charted data.	Yes	Section 4.11, Data Synthesis	The manuscript reports four synthesis stages: descriptive numerical summary, domain-based evidence mapping, thematic

					synthesis, and configurative synthesis. The configurative synthesis integrated evidence into a bounded malaria-SCD syndemic framework.
14	Results, selection of sources of evidence	Provide numbers of sources screened, assessed for eligibility, and included, with reasons for exclusions at each stage, ideally using a flow diagram.	Yes	Section 5.1, Study Selection; Figure 1, PRISMA-ScR Flow Diagram; Supplementary File 8	The manuscript reports records identified, duplicates removed, records screened, records excluded at title and abstract stage, full texts assessed,

					full texts excluded with reasons, and final included sources. Reasons for full-text exclusions are provided in Supplementary File 8.
15	Results, characteristics of sources of evidence	For each source of evidence, present characteristics for which data were charted and provide citations.	Yes	Section 5.2, Characteristics of Included Evidence; Supplementary File 7	The manuscript summarises included evidence by year, country or region, study design, population, data source, disease focus, and evidence domain. Study-level characteristics

					cs are provided in Supplementary File 7.
16	Results, critical appraisal within sources of evidence	If done, present critical appraisal results for included sources.	Not applicable, with explanation	Section 4.12 and Section 6.10	Formal critical appraisal was not conducted because this was a scoping review. Methodological features and limitations of included sources are described narratively and charted where relevant.
17	Results, results of individual sources of evidence	For each included source, present relevant charted data	Yes	Sections 5.3 to 5.9; Supplementary Files 7 and 9	Domain-specific results report charted evidence from

		related to the review questions and objectives.			individual sources across biological interaction, spatial co-risk, clinical convergence, service-mediated visibility, structural vulnerability, programme response, and syndemic theory use. Study-level domain evidence is provided in Supplementary File 9.
18	Results, synthesis of results	Summarise and present charted results as they relate to the review questions	Yes	Sections 5.3 to 5.10	Results are synthesised across seven domains and a cross-domain evidence

		and objectives.			configuration. The synthesis identifies strong evidence for HbAS protection, HbS geography, severe anaemia convergence, SCA under-recognition, hydroxyurea benefit, newborn screening, and mortality under-attribution.
19	Discussion, summary of evidence	Summarise the main results, including how they relate to the review	Yes	Section 6.1, Principal Findings; Section 6.2, Conceptual Contribution	The discussion summarises the major findings and explains how they

		questions and objectives.			support a bounded, serially causal, structurally mediated malaria-SCD syndemic model.
20	Discussion, limitations	Discuss limitations of the scoping review process.	Yes	Section 6.10, Strengths and Limitations	The manuscript discusses heterogeneity of evidence, variable study designs, possible incomplete grey literature capture, limited explicit syndemic studies, absence of meta-analysis, and

					limited evidence on facility readiness, referral networks, geographic accessibility, and complete screening-to-care cascade outcomes.
21	Discussion, conclusions	Provide a general interpretation of results with respect to the review questions and objectives, as well as possible implications or next steps.	Yes	Section 6.11, Conclusion; Section 7, Conclusion	The manuscript concludes that malaria and SCD in SSA are connected through a bounded syndemic pathway involving biological risk, spatial co-risk, clinical convergence

					, service-mediated visibility, structural vulnerability, and programme response. It identifies the need for integrated screening-to-care planning frameworks.
22	Funding	Describe sources of funding for the included sources of evidence and sources of funding for the scoping review, including the role of funders.	Yes	Funding Statement	The manuscript includes a funding statement for the review. The role of funders is stated. If no funding was received, the manuscript states that no specific funding

					supported the review.
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Reference

Tricco AC, Lillie E, Zarin W, O'Brien KK, Colquhoun H, Levac D, et al. PRISMA Extension for Scoping Reviews, PRISMA-ScR: checklist and explanation. *Annals of Internal Medicine*. 2018;169(7):467-473.